

UX and UI Case Studies

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iPod



Palm (PDA)

<http://www.designinginteractions.com/interviews/JeffHawkins>

<http://www.computerhistory.org/revolution/mobile-computing/18/321/2317>



<http://pretotyping.blogspot.pt/2010/08/one-of-my-favorite-pretotype-stories.html>

Mobile Augmented Reality

2003 – First self-contained handheld Augmented Reality system (Mobile AR)



D. Wagner and D. Schmalstieg, "First steps towards handheld augmented reality," *Seventh IEEE International Symposium on Wearable Computers, 2003. Proceedings.*, 2003, pp. 127-135

User eXperience (UX)

Emphasizes the motivations and experiences of users.

Experiences are created as a result of an active interpretative process that is shaped by mood, context, attention, goals, and attitudes, among other factors.

UX Goals

Desirable aspects

Satisfying

Enjoyable

Engaging

Pleasurable

Exciting

Entertaining

Helpful

Motivating

Challenging

Enhancing sociability

Supporting creativity

Cognitively stimulating

Fun

Provocative

Surprising

Rewarding

Emotionally fulfilling

Undesirable aspects

Boring

Frustrating

Making one feel guilty

Annoying

Childish

Unpleasant

Patronizing

Making one feel stupid

Cutesy

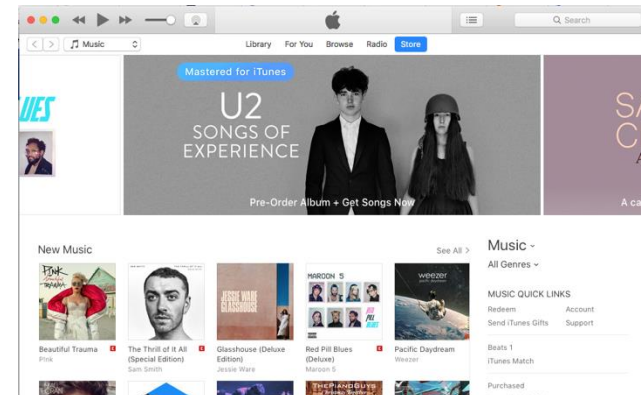
Gimmicky

Preece, J, Sharp, H, Rogers, Y. Interaction Design: Beyond Human-Computer Interaction, 5th Edition. 2019. Wiley

User eXperience (UX)

“Encompasses all aspects of the end-user's interaction with the company, its services, and its products.”

<https://www.nngroup.com/articles/definition-user-experience/>



Usability vs UX

“According to the definition of usability, it is a quality attribute of the UI [...] user experience is an even broader concept.”

<https://www.nngroup.com/articles/definition-user-experience/>

UX

Example: Gamer Experience

FUNCTIONAL

Consequences

Ease of Control

Progress Feedback

Audiovisual Appeal

Goals and Rules

Challenge



PSYCHOSOCIAL

Consequences

Mastery

Curiosity

Immersion

Autonomy

Meaning

UI

Examples and Case Studies

How fast are things perceived?

< ~20ms (1/50 sec) discrete images/flashes
merge into continuous perception

- Image you are looking at flickers 60 times per second
- Differences in peripheral vision



How fast are things perceived?

< ~20ms (1/50 sec) Displays update at 50-75hz

Pretty much never have to be faster than this for user response!

Get >120 million instructions per core (@3Ghz)

(High end GPU theoretically as high as 26 billion instr.)

- You can do a lot with that
- (First GUIs had ~20K)
- Not enough? Twice as many cores in 2-3yrs...

How fast are things perceived?

- < ~100ms seems like “instant response”
 - Hard to tell response times below this apart
 - Upper range of eye saccades

Discrete images drop into steps
instead of apparent motion

How fast are things perceived?

100ms (1/10 sec)

Except some animation, most things don't need to be faster than this

- Typical target “cycle time”
- 600 million+ machine instructions, ...

How fast are things perceived?

- < 1-2 seconds typically “good response time”
 - Similar times in conversational turn taking protocols
 - Longer delays ~5 sec have to say something to keep conversation alive

(Note: numbers fuzzier as we go out)

How fast are things perceived?

- > 10-15 sec is typically “bad response time”
 - STM decay effects

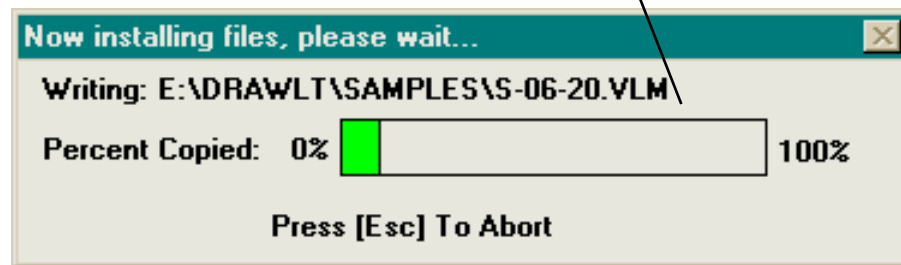
A little about response times

Good vs. bad response time
is very dependent on expectation

If you can't meet the goals, manipulate user expectations!

Consistency of response is very important
– Can be more important than time

Multiple files being copied,
but feedback is file by file.



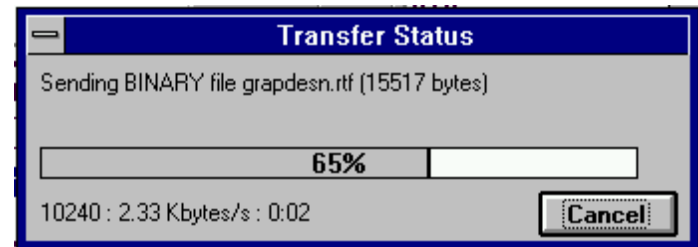
Drawing Board LT

Dealing with long delays

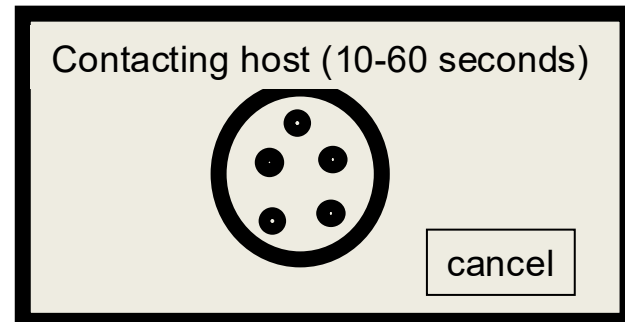
- Cursors
 - for short transactions



- Percent done dialogs
 - time left
 - estimated time



- Random
 - for unknown times

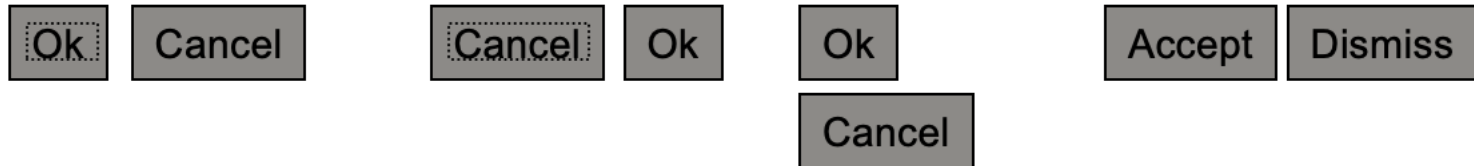


Be consistent

Consistent syntax of input

Consistent language and graphics

- same visual appearance across the system (e.g. widgets)
- same information/controls in same location on all windows



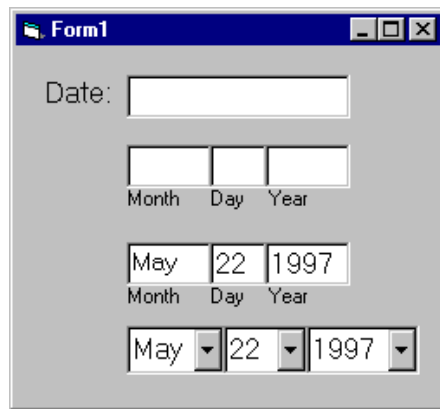
Consistent effects

- commands, actions have same effect in equivalent situations
 - predictability

Deal with errors in a positive manner

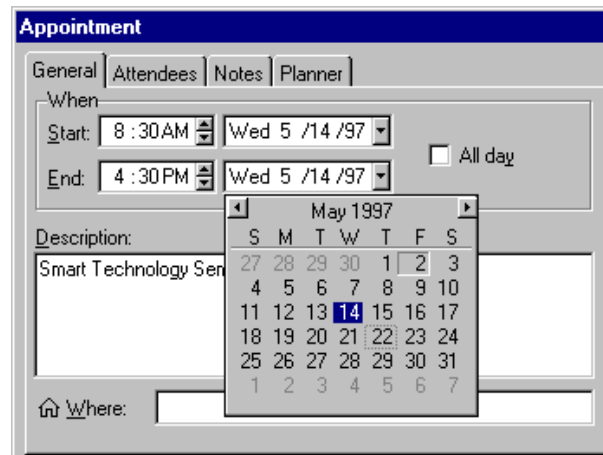
Prevent errors

- try to make errors impossible
- widgets where you can only enter legal data



The 'Form1' window displays three different date input methods:

- A single text box labeled 'Date:'.
- Three separate text boxes for 'Month', 'Day', and 'Year'.
- A dropdown menu for 'Month' (showing 'May'), a text box for 'Day' (showing '22'), and a dropdown menu for 'Year' (showing '1997').
- A combined dropdown menu for 'Month' (showing 'May'), a dropdown menu for 'Day' (showing '22'), and a dropdown menu for 'Year' (showing '1997').



The 'Appointment' window shows a 'General' tab with the following fields:

- 'When' section: 'Start' (8:30 AM) and 'End' (4:30 PM) time pickers, and a date picker (Wed 5 /14 /97).
- 'All day' checkbox.
- 'Description:' text area (containing 'Smart Technology Sen').
- 'Where:' text box.

A calendar popup for May 1997 is displayed, showing the date 14 (Wednesday) selected.

Provide reasonableness checks on input data

- on entering order for office supplies
 - 5000 pencils is an unusually large order. Do you really want to order that many?